

School of Information Science and Engineering

International Crash Course and Practical Training Week

Lanzhou University, Gansu, China – Mon-Friday, July 20-24, 2026

Network and OS Foundations for Safe Robotic Actuation: *Latency, Reliability, and Event-Driven Control in Industrial Hair-Cutting Robots*

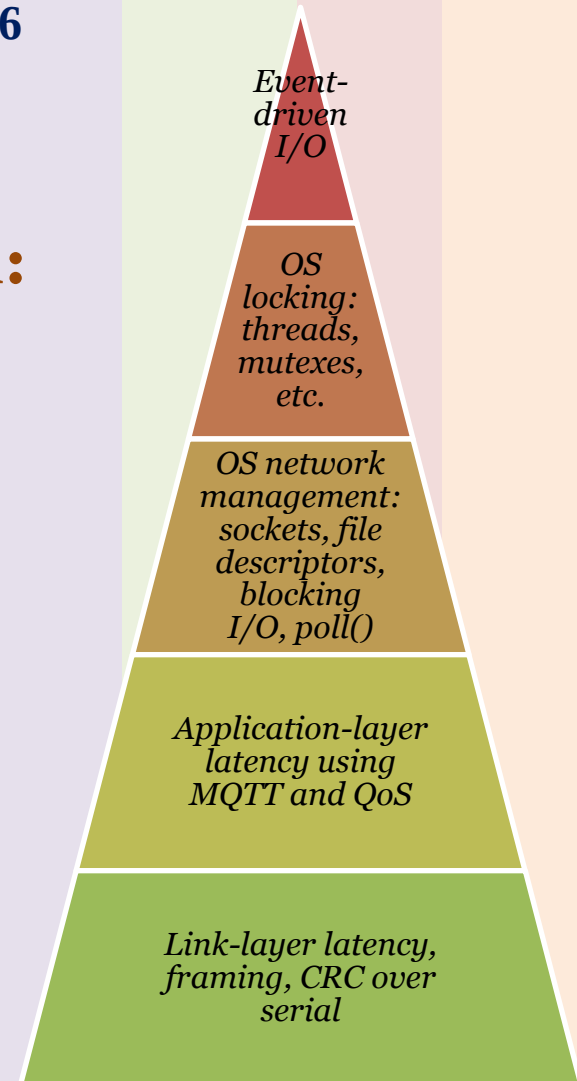
Audience Level: Advanced Undergraduate / Early Post-Graduate

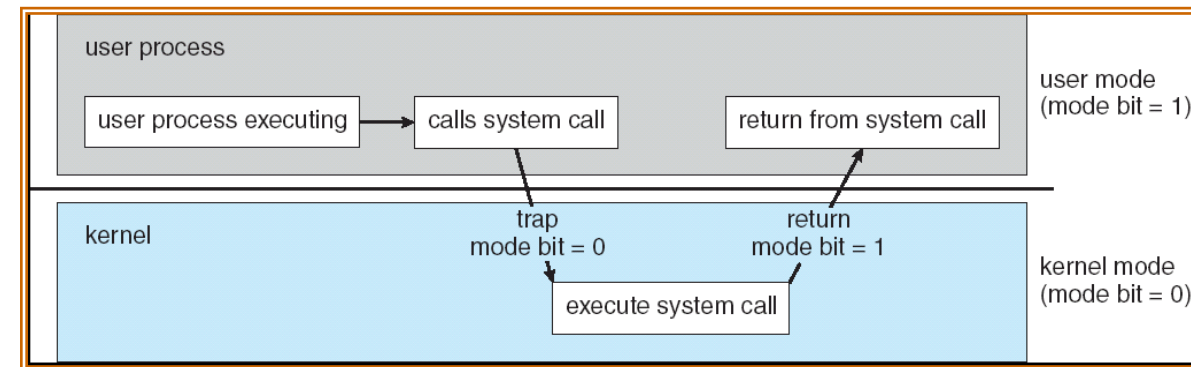
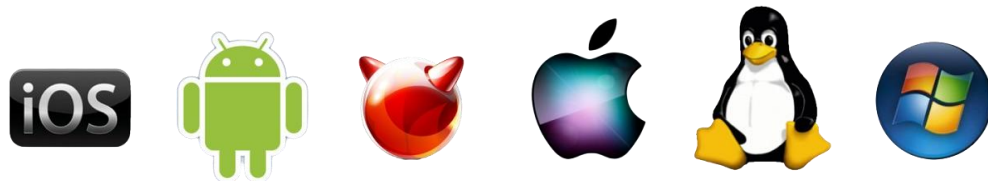
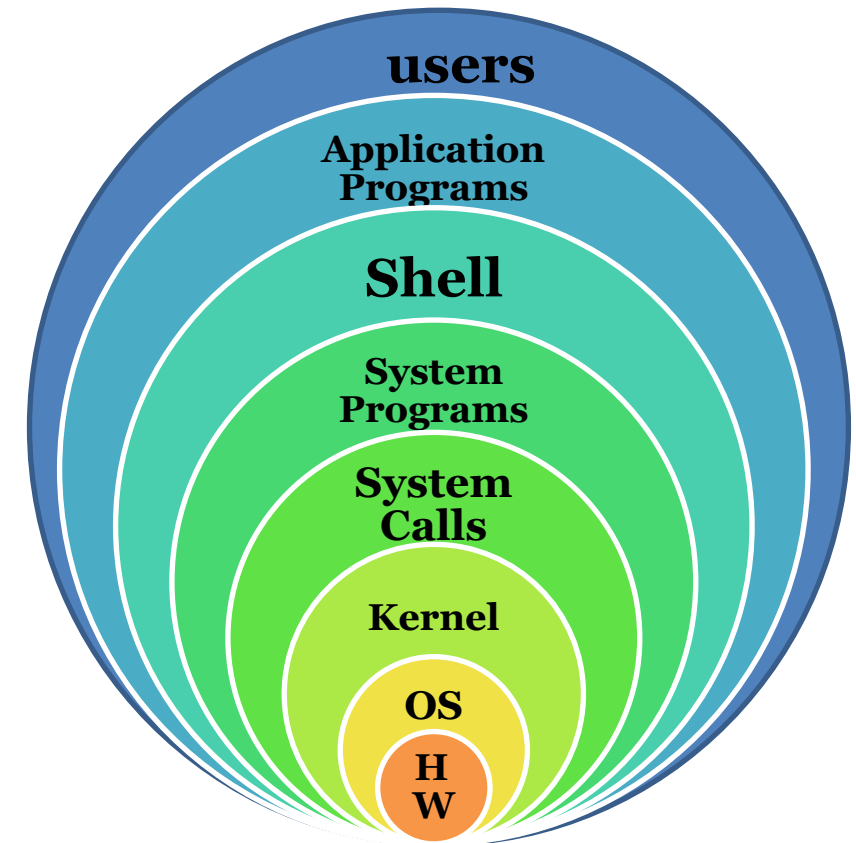
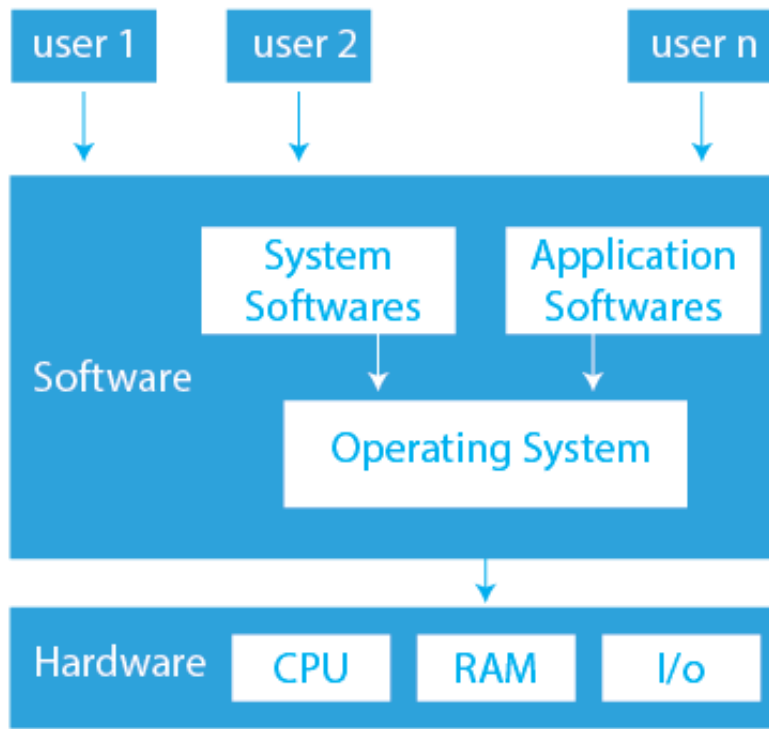
Prepared and Presented by

Prof. Dr. Mohammed Aquil Mirza

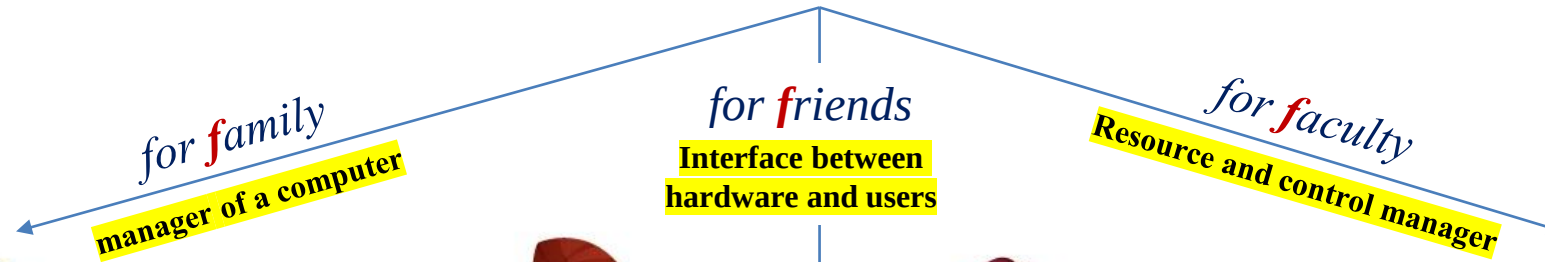
*Fellow IESE (UK), Fellow IETE (India), Fellow HKIE (Hong Kong SAR),
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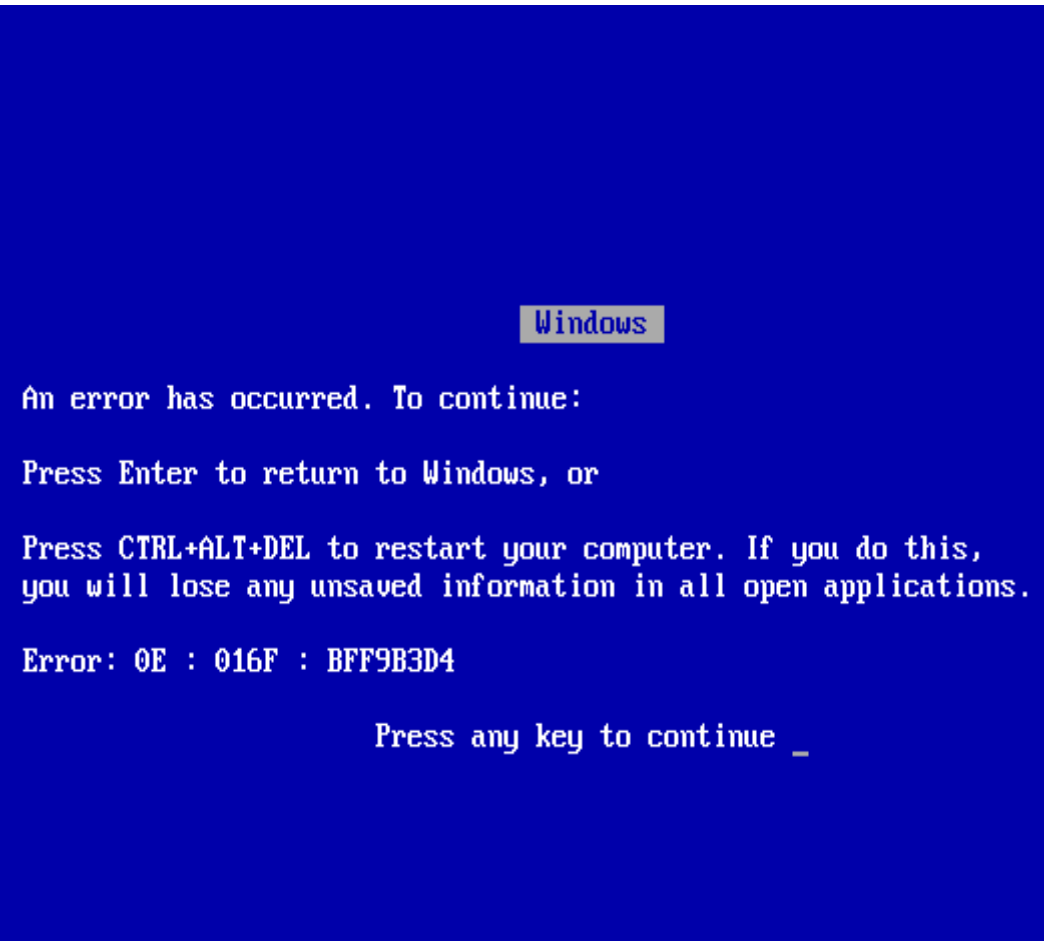


OS DEFINITION



Failures of Operating Systems

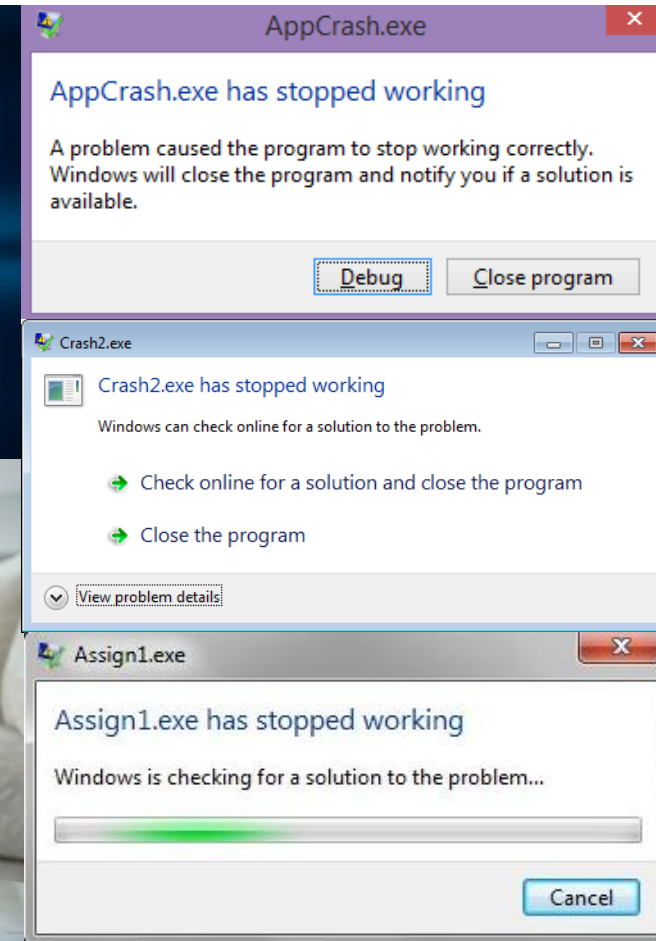
Example 1: Blue Screen of Death (BSOD)



Example 2: Computer Freezing



Example 3: Application Crashes



Failures of Operating System?

The Intercept

While the incident involving Tesla cars may point to potential failures within the vehicle's operating system or software, it is essential to consider the broader context. Factors such as software bugs, sensor performance, user engagement, and environmental conditions all contribute to the safe operation of autonomous vehicles.

DONATE



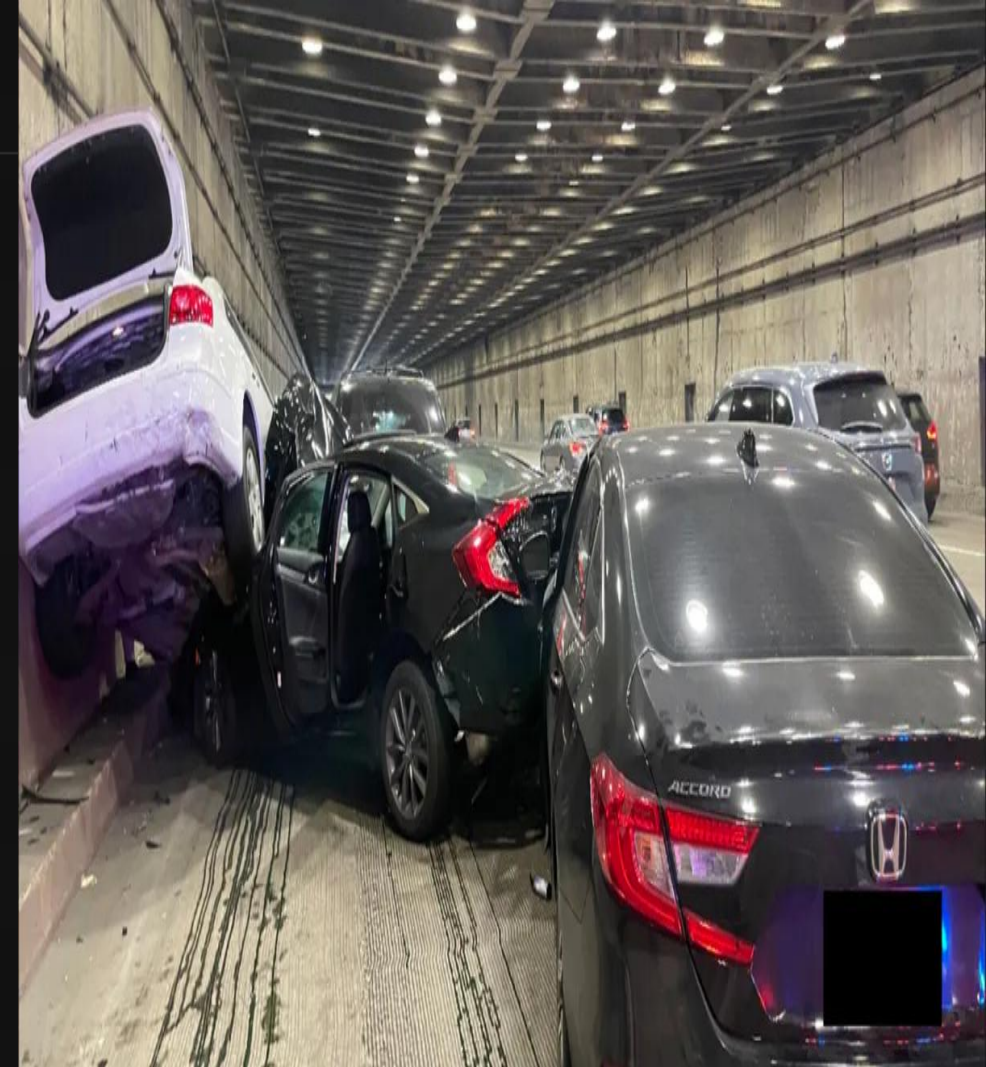
EXCLUSIVE: SURVEILLANCE FOOTAGE OF TESLA CRASH ON SF'S BAY BRIDGE HOURS AFTER ELON MUSK ANNOUNCES "SELF-DRIVING" FEATURE

Musk has said Tesla's problematic autopilot features are "really the difference between Tesla being worth a lot of money or worth basically zero."

Ken Klippenstein

January 10 2023, 11:22 a.m.

Share



Interrupts & System Calls



Before the Crime

A. Police patrol to report crime (Interrupt)



During the Crime

B. Police stopping crime (Interrupt)



After the Crime

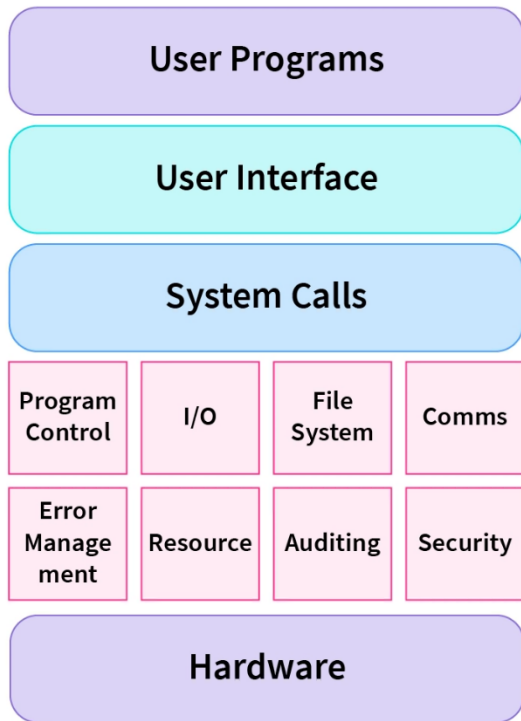
C. Police after event/crime (Interrupt)



System calls = a mechanism to access OS Services

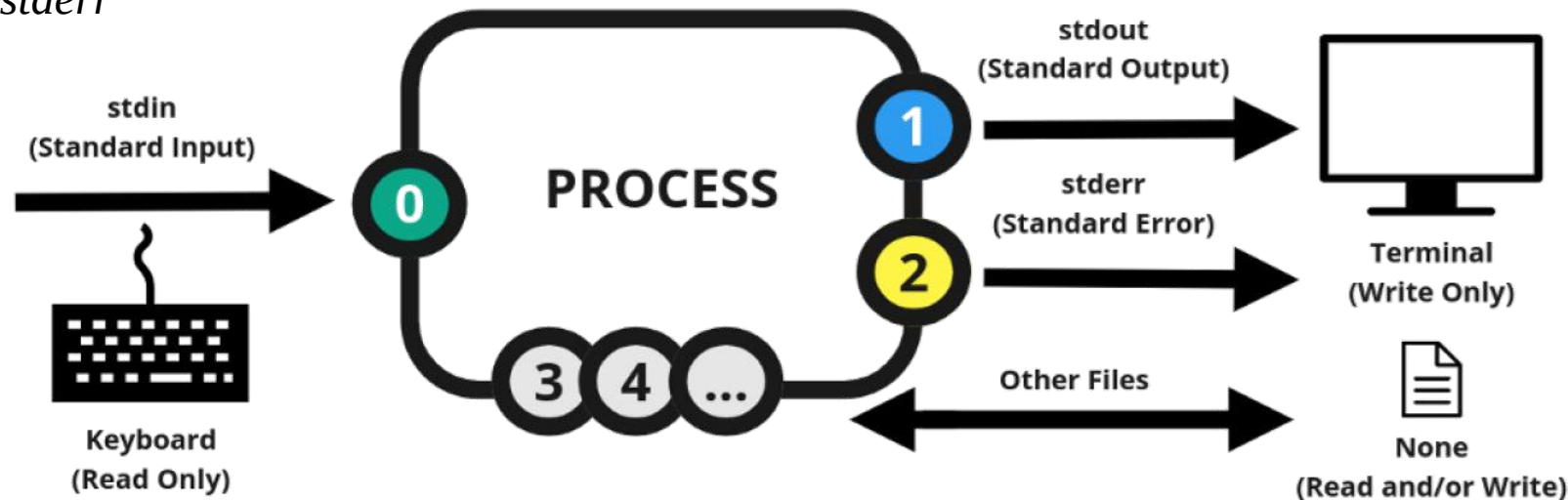
read(), write(), open() system call

Introduction to System Call



file descriptor (fd)

- 0** – *stdin* (keyboard)
- 1** – *stdout* (Monitor)
- 2** – *stderr*



Named vs Unnamed Pipe

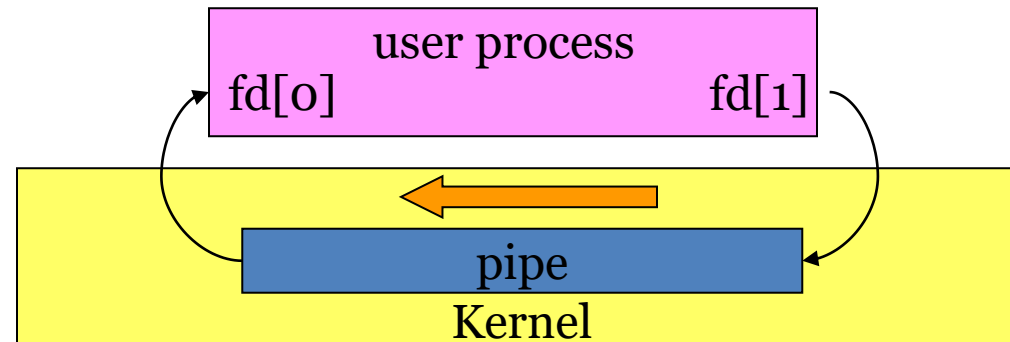
A **named pipe** is actually a **file**.

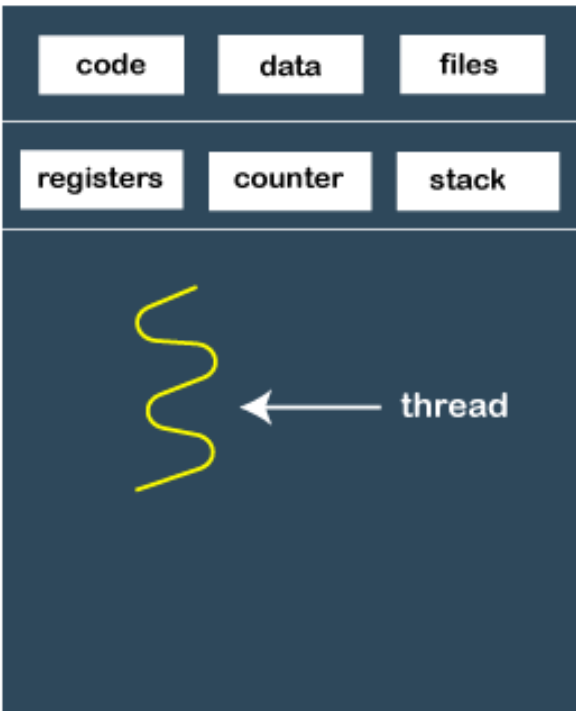
A named pipe is created by creating a **special file**, via the system call **mkfifo()**.

After a named pipe is created, any two processes knowing the name of the file will first **open()** the pipe like a file. They can then communicate by writing to and reading from it, using **write()** and **read()** accordingly like a file.

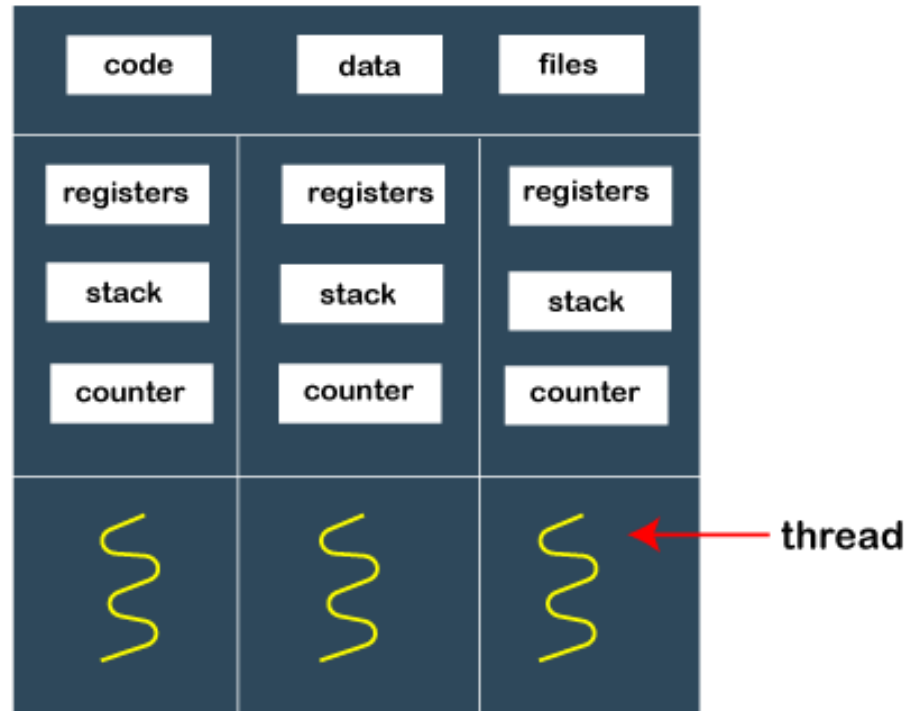
The pipe **remains** even after both processes have completed.

It is a good programming practice to **create** the named pipe and then **delete** the named pipe inside the program at the end, via **unlink()** system call.





Single-threaded process



Multi-threaded process

Processes

Threads

- Top-level execution container
- Separate memory space

- Runs inside a process
- Shared memory space